

DVCon India 2026 Design Contest — Stage 3A

Submission

Team ID: 166 (Byte Silicon)

Project: GatE-V — Task-Aware Object Detection on VEGA AS1061 RISC-V + Kintex-7 FPGA

1 Executive Summary

This archive contains Team 166’s official Stage 3A submission for the DVCon India 2026 Design Contest. The submission integrates our custom **512-MAC Weight-Stationary Systolic Array Accelerator** (GatE-V) into the **C-DAC VEGA AS1061 RISC-V SoC** environment.

Key Performance Highlights

- **Matrix Engine:** 32×16 (512 MAC cells) executing INT8 weight-stationary matrix operations.
- **Integrated SoC Clock:** 50 MHz system clock delivering 51.2 GOPS peak throughput.
- **Resource Utilization (Kintex-7 xc7k325tffg900-2):**
 - Integrated SoC Total: 88,054 Slice LUTs (43.21%), 594 DSP48E1 (70.71%), 122 BRAMs (13.70%).
 - GatE-V Accelerator Alone: 5,356 Slice LUTs (2.63%), 544 DSP48E1 (64.76%).
- **Verification:** Reproducible bare-metal RISC-V self-test firmware (gatev_firmware) running on the VEGA AS1061 CPU.

2 Directory Structure

```
DVCon_India_2026_DC_Stage3A_166/
|-- Document/
|   |-- DVCon_India_2026_DC_Stage3A_Report_166.pdf  <- Stage 3A IEEE Report
|   |-- simulation_logs_summary.txt                <- Simulation summary
|   '-- figures/                                    <- Figures & waveform PNGs
|-- Readme/
|   |-- README.md                                  <- Execution instructions (Markdown)
|   '-- Readme.pdf                                  <- Execution instructions (PDF)
'-- ENV/
    |-- Application/                                <- Bare-metal C application firmware (gatev_firmware)
    |-- DVCon_SoC_SRC/                              <- C-DAC VEGA AS1061 RISC-V SoC + GatE-V IP
    |   |-- ACCELERATOR_IP/                          <- GatE-V 512-MAC Systolic Array RTL
    |   |-- TOP/                                       <- VEGA AS1061 SoC Top Integration
    |   |-- TB/                                        <- Testbench environment
    |   '-- MEMORY_IP/                               <- Boot ROM models (rom_32KB_axi)
    |-- DVCon_SoC_XDC/                              <- Kintex-7 constraint files
```

```

|-- GEN_BIT_OUT/          <- Boot ROM & bitstream configuration
|-- RUN/                  <- Automated simulation scripts (run.sh) & rom_32KB_axi
|-- TCL/                  <- Vivado synthesis scripts (DVCon_SYN.tcl)
'-- VIVADO_PROJECT/       <- Vivado synthesis project directory

```

3 How to Run Simulation & Verification

To run the automated C-DAC simulation testbench:

```

cd ENV/RUN
./run.sh

```

Or for GUI simulation in ModelSim/QuestaSim:

```

vsim -do simulate.do

```

Building the Bare-Metal Self-Test Firmware

To rebuild the RISC-V self-test firmware and boot ROM:

```

cd ENV/Application/gatev_firmware
make clean && make
python3 bin2mif.py build/gatev_test.bin rom_32KB_axi.mif
cp rom_32KB_axi.mif ../../RUN/rom_32KB_axi.mif

```

4 Hardware Memory Map (Base Address: 0x2004_0000)

Offset	Register Name	Access	Description
0x00	CTRL	W	Bit 0: Start computation
0x04	STATUS	R	Bit 0: Done flag
0x08	TASK_ID	R/W	Algorithmic Task ID
0x0C	MODE	R/W	Configuration mode
0x10	IMG_BASE_ADDR	R/W	SRAM Base Address for activation image
0x14	WT_BASE_ADDR	R/W	SRAM Base Address for layer weights
0x18	OUT_BASE_ADDR	R/W	SRAM Base Address for convolution output
0x1C	TILES_NUM	R/W	Total convolution tiles to process
0x30	BURST_TYPE	R/W	AXI burst type (INCR)
0x34	BURST_LEN	R/W	AXI burst length (64 beats)
0x40--0x54	PERF_*	R	Performance counters